

ADRIAN DENDORFER

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Education

M.Sc Neuroengineering, Technical University of Munich (TUM)

10/2023 – Present

Current grade in our mandatories & expected grade: 1.1. Best in class.

Highlights:

- Student representative. Co-organized the MSNE summer school 2024.
- Completed a [PhD course on probabilistic AI](#) from the Norwegian University of Science and Technology.
- Received an [honorable mention](#) for an essay from the International Neuroethics Society.

Currently: Thesis at [MRC Cognition and Brain Sciences Unit at the University of Cambridge](#):

- Thesis: 'Mapping the Multi-Constraint Landscape of Mammalian Brain Network Architecture'
- Manuscript: 'Benchmarking Network Comparison Metrics for Generative Models in Network Neuroscience' (final steps of preparation)
- Fellow in the [Alignment Fellowship](#) of the Cambridge AI Safety Hub.

Exchange Semester, Georgia Institute of Technology

01/2023 – 05/2023

Final Grade: GPA 4.0 (18 credit hours).

B.Sc Electrical Engineering and Information Technology, TUM

10/2020 – 03/2024

Final Grade: 1.4 (180 ETCS, approx. top 5%).

B.Sc Engineering Science, TUM

10/2019 – 11/2023

Final Grade: 2.1 (210 ETCS, approx. top 35%) Leader for a resort of the student body of the Munich School of Engineering.

Research Experience

IBM Research Zurich & TUM.ai

10/2024 – 04/2025

Researched number handling in LLMs by adapting loss functions.

Heidelberger Institute of Global Health, Heidelberg University & Munich Data Science Institute, TUM

10/2024 – 02/2025

TUM-DI lab. Worked on infection detection from wearable's data. Used GANs for time-series imputation, and bi-directional LSTM-autoencoders for anomaly detection.

Department of Public Health, University of Cambridge

08/2024 – 11/2024

Researched the analysis of pulse wave morphology in photoplethysmography recordings (the green light in wearables), 40h/week.

Worked from Cambridge and remotely.

AI in Medicine Lab, TUM

10/2023 – 06/2024

Benchmarked algorithms for the prediction of 'brain age' (a biomarker for cognitive fitness) from EEG data utilizing both conventional signal processing and deep learning, 12h/week.

MAP Lab, Georgia Institute of Technology

01/2023 – 10/2023

Researched how to predict a person's spatial orientation from their brain imaging data (fMRI and fNIRS) to be used in real-time neurostimulation feedback loops. 12h/week.

Worked from Atlanta and remotely.

Little Devices Lab, Massachusetts Institute of Technology (MIT)

04/2022 – 07/2022

Developed magnetic switches for a paper-based and modular microfluidic toolkit.

Worked remotely due to pandemic.

Institute for Cognitive Systems, TUM

07/2022 – 10/2022

Worked on a device to help a patient with multiple sclerosis to balance. Accepted for [NER 2023](#).

Published Work

Wallner, J., Berbuir, S., Birner, L., Dendorfer, A., Panthi, B., Rahnsch, B., Muma, J., Munga, S., Obor, D., Bärnighausen, T. and Barteit, S., 2026. Overcoming Data Loss in Wearable Disease Detection with GAN-Based Imputation. **NPJ Digital Medicine**. DOI: <https://doi.org/10.1038/s41746-026-02518-4>

Dendorfer, A., and Charlton, P.H., 2025. Wrist Photoplethysmography Pulse Waves: Morphological Classes and Physiological Influences. In **Eng. Proc.** DOI: <https://doi.org/10.3390/ECSA-12-26556>.

Zausinger J., Pennig L., Kozina A., Sdahl S., Sikora J., **Dendorfer A.**, Kuznetsov T., Hagog M., Wiedemann N., Chlodny K., Limbach V., Ketteler A., Prein T., Singh, V. M., Danziger, M., Born, J., 2025. Regress, Don't Guess - A Regression-like Loss on Number Tokens for Language Models. In **2025 International Conference on Machine Learning (ICML)**. DOI: <https://doi.org/10.48550/arXiv.2411.02083>.

Pilger, L., Berberich, N., Paredes-Acuña, N., **Dendorfer, A.**, Guadarrama-Olvera, J. R., Bergner, F., ... & Cheng, G. (2023, April). Human-centered design of a vibrotactile sensory substitution belt for feet somatosensation in a patient with multiple sclerosis. In **11th International IEEE/EMBS Conference on Neural Engineering (NER)** (pp. 1-4). IEEE. DOI: <https://doi.org/10.1109/NER52421.2023.10123871>.

Non-Research Work Experience

Intern, DocumentAI Worked on business design for the early-stage startup (10 h/week).	09/2022 – 12/2022
Consultation Project, Messe München Developing sustainable (ESG) business ideas for Messe München (12 h/week).	10/2021 – 02/2022
Tutor for Electrical Engineering, Chair of Signal Processing, TUM	10/2021 – 02/2022
Project Lead and Engineer, social startup reStove Led a team of twelve through the development of frugal biogas plants for use in our partner-NGO's network (15-20 h/week). Highlights: Developed prototypes in India and Germany; created and sustained partnerships with multiple companies, research labs and NGOs in both countries.	05/2020 – 04/2022
Intern, V-R-Robotics Experimented with robot navigation algorithms (6 weeks * 40 h/week).	09/2020 – 10/2020

Extracurricular Activities

Recruiting Lead, TUM.ai Organized recruiting, leading 17 students. We nearly doubled the total number of applications compared to the semester before (by 90%, to 418); the female/male ratio of applicants increased by 185%.	01/2025 – 05/2025
Organizer and Judge of the Impact Investing Competition, Manage and More Organized a competition for impact-orientated social startups worldwide (10.000€ prize).	10/2023 – 02/2024
Member of the “Board of Directors”, Enactus Munich Co-management of Enactus Munich, a student initiative that aims to develop entrepreneurial solutions to global challenges.	02/2021 – 05/2022
Youth Parliamentarian and Treasurer, Youth Parliament of my hometown	09/2018 – 04/2021

Scholarships

I highly value the learning opportunities and support provided by the [Max Weber Program](#), [Manage and More](#), and the [Elite Graduate Program](#) (Elite Network of Bavaria). Furthermore, I am grateful for the financial support from the [Deutschlandstipendium](#), [Erasmus+](#), the offered support by [PROMOS](#) (declined), and the [Heinrich and Lotte Mühlfenzl Foundation](#) for parts of my academic journey.